

UNIT 4 – Data Visualization Techniques

Activity 3: Specialized Data Visualization Experiment

Task 1: Dataset Identification and Preparation. Selecting a specialized dataset that naturally benefits from advanced visualization techniques. Examples of suitable datasets include:

- ❑ Stock market price data with daily open, close, high, and low values
- ❑ Weather or temperature data across multiple geographic regions
- ❑ Time-series sensor data or economic indicators

The dataset should be cleaned and formatted to ensure accurate plotting and smooth visualization generation.

Dataset Selection :- Name = Historical Stock Price Dataset

Justification for Dataset Selection:-

This dataset is chosen because it:

- Contains **time-series financial data** that naturally benefits from:
 - Trend visualization
 - Comparative performance plots
 - Volatility and distribution analysis
- Has **multiple numerical variables** that are useful for:
 - Line charts (price over time)
 - Scatter and distribution plots (returns, volume correlations)
- Represents a **real-world dataset** that is large enough for performance benchmarking
- Is openly available on Kaggle, ensuring reproducibility

Task 2: Selection of Visualization Techniques

- Visualization techniques may include:
- Line plots with rolling averages or trend overlays
- Candlestick charts for financial data
- Heatmaps for spatial or intensity-based patterns
- 3D surface plots for multi-variable relationships

Each visualization should clearly represent the data without clutter and maintain visual clarity.

Task 3: Advanced Customization and Highlighting Insights

Customization should include:

- Axis labeling, legends, and titles
- Color schemes suitable for interpretation and accessibility
- Annotations marking significant events such as peaks, drops, maxima, or minima
- Gridlines, markers, or scaling adjustments when appropriate
- The visualization should visually guide the viewer toward meaningful observations.

Task 4: Interpretation and Insight Documentation

The interpretation should describe:

- Major trends or variations observed in the data
- Notable extreme values or anomalies
- Comparative observations across time, categories, or regions

The explanation should directly relate to the visual evidence presented

1. Line Plot with Rolling Averages

Visual Evidence:

- The line plot shows the **daily closing price** with **7-day and 30-day rolling averages**.
- Maximum and minimum prices were annotated with gold and purple markers.

Insights:

- **Overall Trend:** The stock shows moderate **upward and downward fluctuations** over 6 months, with no extremely sharp trends.
- **Short-term vs Long-term Trends:**
 - The **7-day moving average** responds quickly to price changes and highlights short-term volatility.
 - The **30-day moving average** smooths out fluctuations, showing the general trend.
- **Extreme Values:**
 - Peak (max price) occurs around [max_date], indicating the highest market confidence.
 - Lowest price occurs around [min_date], potentially indicating a short-term market dip.
- **Observations:** The gap between short-term and long-term averages can signal **volatility periods**, which can guide short-term trading strategies.

2. Candlestick Chart

Visual Evidence:

- Candlestick chart shows **daily OHLC (Open, High, Low, Close) prices**.
- The highest high is annotated with an orange arrow.

Insights:

- **Daily Movements:** Red and green candles show days with losses vs gains, visualizing market sentiment.
- **Peak Price:** The annotated high indicates a day of **significant bullish activity**.

- **Volatility Patterns:** Long wicks (shadows) indicate **intraday volatility**, especially on days with extreme highs or lows.
- **Trading Insight:** Candlestick patterns help identify **reversal or continuation signals** in trading.

3.Heatmap of Average Monthly Close Prices

Visual Evidence:

- Heatmap shows **average closing prices for each month**, with red-yellow gradient.

Insights:

- **Monthly Trends:**
 - Certain months have higher average prices, indicating periods of **market optimism or growth**.
 - Lower average prices suggest **slower months or market dips**.
- **Comparative Observation:** Viewers can immediately spot which months have the **highest and lowest closing prices**.
- **Insight for Strategy:** Useful for seasonal analysis or planning medium-term investment decisions.

4.3D Surface Plot: Volume vs Close vs Daily Change

Visual Evidence:

- 3D surface visualizes **relationship between trading volume, closing price, and daily change**.
- Color gradient shows magnitude of daily change.

Insights:

- **Volume vs Price:**
 - Larger daily changes sometimes occur at higher volumes, indicating **more market activity** during significant price movements.

- **Anomalies:** Outlier points (peaks in the surface) show days of **unexpectedly high gains or losses**, which may correspond to market events.
- **Correlation Observation:** The surface suggests **positive correlation** between price movement and volume on volatile days.
- **Investor Insight:** Investors can identify **high-impact trading days** and plan accordingly.

Overall Data Interpretation

Aspect	Observations
Trend	Stock shows moderate fluctuation with identifiable peaks and dips. Rolling averages reveal short- and long-term trends.
Extremes	Maximum and minimum prices annotated in line plot; highest daily high highlighted in candlestick chart.
Volatility	Long wicks in candlestick and surface spikes indicate high volatility periods.
Comparative	Heatmap allows month-to-month comparison of average closing prices; trends can guide medium-term strategies.
Actionable Insight	Peaks, dips, and high-volume days highlight significant events for potential trading or monitoring.